

Samalka Anandagoda, Ph.D.

📍 Chester, Maryland
☎ +1 (864) 624 6642
✉ samalka.piy@gmail.com

LinkedIn: <https://www.linkedin.com/in/samalka-anandagoda/>

GitHub: <https://github.com/samalkapiy>

SUMMARY

- Data scientist and computational researcher with 8+ years of experience analyzing large data sets, building statistical and computational models that resulted in 6+ peer reviewed publications (<https://samalkapiy.github.io/publications/>) and a software release (SNuGGY).
- Experienced in data preprocessing, feature engineering, hyperparameter tuning, model training and model evaluation for Machine Learning projects such as credit card fraud detection (full list of projects: <https://samalkapiy.github.io/projects/>).

TECHNICAL PROJECTS

- Credit Card Fraud Detection model built using XGBoost achieving an accuracy of 99.99%. Net monthly savings from utilizing the model: 58,638 dollars.
- Space-X Launch success prediction model with XGBoost (cross validation score: 0.89) leading to a 51 million dollars in savings per launch by accurately predicting the outcome of the booster rocket.
Full list containing the slides, codes used for the projects found at my website: <https://samalkapiy.github.io/projects/>

LICENSES AND CERTIFICATIONS

IBM Data Science Professional Certificate.

Issued : July 1st 2025, Credential ID: [A7TEPLV81V4K](#)

RELEVANT RESEARCH AND WORK EXPERIENCE

Aug 2024 - Present	Visiting Assistant Professor of Physics -Conducted quantitative analysis on ~20 GB of astrophysical data using pandas and NumPy, building computational models, optimizing with scipy to extract insights to answer key questions in research.	Washington College, Maryland
May 2022 - Aug 2023	Research Assistant Developed and debugged a Monte Carlo particle transport simulation code written in C++. The code incorporates advanced computational algorithms and physics-based simulations and run on High Performance Computing cluster (skills: job scheduling, bash scripting).	Los Alamos National Laboratory, NM
Aug 2016- Aug2023/	Astrophysics Researcher/ Volunteer research fellow -Development of a Github hosted software release (https://github.com/adesai90/SNuGGY) -Building computational models to answer astrophysical research questions. This project utilized programming skills such as python, parameter fitting using maximum likelihood estimations, writing SQL queries to obtain data from astrophysical catalogs, data visualization and model building.	Clemson University, SC
Aug 2023- Aug 2024		

SOFTWARE RELEASES

SNUGGY: Simulation package mapping ~ 1 million source locations in the Milky Way galaxy utilizing a 2D probability distribution function (PDF) to understand particle interaction processes.

SKILLS

Tools	Python (PyTorch, SciKit, pandas, matplotlib, numpy, scipy, seaborn), C++, SQL, Jupyter Notebooks, Git
Concepts	Statistics, Machine Learning, Clustering algorithms, Regression and Classification, Differential equations, Monte Carlo method. Git version control, High performance computing.

EDUCATION

Aug 2019 - Aug 2023	Ph.D. in Astronomy and Astrophysics	Clemson University, SC
Aug 2016 - Aug 2019	MS in Astronomy and Astrophysics	Clemson University, SC
Nov 2011 - Nov 2015	Bachelor of Science in Physics Major: Physics, Minor: Computer Science and Mathematics	University of Peradeniya, Sri Lanka